



StudyMate - Rule-Based Study Assistant



Project Overview

StudyMate is a Rule-Based Study Assistant developed using Python and Tkinter. It is designed to answer basic academic questions related to computer science subjects using predefined rules and keyword matching.

The application provides a simple graphical user interface (GUI) where users can interact with the chatbot by asking study-related questions. It is intended to help beginners quickly revise important concepts in programming and computer science.



Features

- User-friendly graphical interface using Tkinter
 - Rule-based chatbot implementation
 - Answers frequently asked study-related questions
 - Supports greetings and basic conversation
 - Handles unknown questions with fallback responses
 - Clear Chat functionality
 - Responsive chat interface with auto-scrolling
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Technologies Used

- Python 3.x
 - Tkinter
 - Object-Oriented Programming (OOP)
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Project Structure

```
StudyMate/  
|  
├─ main.py  
├─ gui.py  
├─ chatbot.py  
├─ responses.py  
├─ utils.py  
└─ README.md
```

How to Run

1. Install Python 3.x.
2. Download or clone the project.
3. Open the project folder in Visual Studio Code.
4. Run the following command:

```
python main.py
```

The StudyMate GUI window will open.

Sample Questions

Try asking questions like:

- What is Python?
 - Explain Object Oriented Programming.
 - What is SQL?
 - Difference between List and Tuple.
 - What is Machine Learning?
 - What is Data Science?
 - Hello
 - Bye
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Learning Objectives

This project demonstrates:

- Rule-Based Chatbot Design
 - Python Programming
 - GUI Development with Tkinter
 - Modular Programming
 - Object-Oriented Programming
 - Event-Driven Programming
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Future Enhancements

- AI-powered responses using Large Language Models (LLMs)
- Voice input and speech output
- Student login system
- Quiz mode

- Subject-wise learning modules
 - Database integration
 - Performance tracking
 - Dark mode interface
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Author

Saurav Shukla

B.Tech (Computer Science & Engineering)

License

This project is developed for educational and learning purposes.